

Nishan Shetty MS, MBA

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EDUCATION

Master of Business Administration , Population Health Management - Ottawa University	May 2024
Master of Science , Industrial Engineering - Arizona State University	May 2019

CERTIFICATIONS

NVIDIA-Certified Associate, Generative AI LLMs | **AWS**, Certified Machine Learning - Specialty | **Databricks**, Machine Learning Engineer - Associate | **Microsoft**, Certified Azure Fundamentals | **Microsoft**, Certified Azure AI Engineer Associate

SUMMARY

AI Engineering Manager with 8+ years leading high-performing teams across data curation, model fine-tuning, and agentic AI development. Proven ability to define strategic AI vision, balance short and long-term delivery goals, mentor senior engineers, and serve as the technical liaison between engineering, leadership, and cross-functional stakeholders. Stays 40-60% hands-on across the full AI lifecycle: data pipelines, RLHF and instruction tuning, and production-grade multi-agent systems on AWS and Azure.

SKILLS

Leadership & Strategy: Team management (6-8 direct reports) | Technical roadmap | Code review | Design review | Senior engineer mentorship | Cross-functional liaison | Stakeholder communication | Performance management | Agile delivery

Agentic AI & LLMs: LangChain | LangGraph | Multi-agent systems | Tool-use | Agent memory | Task planning | RLHF | Instruction tuning | LoRA | Prompt engineering | OpenAI APIs | AWS Bedrock | HuggingFace | GPT | Claude | Llama

NLP & Classical ML: spaCy | BERT | word2vec | Flair | Entity extraction | Text classification | Embeddings | Time Series (ARIMA, LSTM) | Clustering | Causal inference | A/B testing | Uplift modeling

MLOps & Data: MLflow | SageMaker | Azure ML | Docker | CI/CD | Airflow | Lambda | Spark | Databricks | Snowflake | Kafka | Drift detection | Retraining pipelines | HIPAA compliance

WORK EXPERIENCE

Senior AI Engineer	December 2018 - Present
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Southwest Medical Imaging | Arizona

Team Leadership & Strategic Vision

- Managed a team of 6 data scientists, ML engineers and analysts across three parallel workstreams: data curation, model fine-tuning, and agentic AI system development. Defined the technical roadmap, balanced short-term delivery with long-term AI capability building, and conducted regular code and design reviews enforcing production readiness criteria.
- Mentored and guided senior engineers through day-to-day technical direction, design reviews, and structured feedback, fostering a collaborative and innovative culture. Acted as the liaison between the AI team, executive leadership, product, and clinical stakeholders, communicating project status and AI investment trade-offs clearly.

Agentic AI & GenAI Development

- Clinical staff were manually processing high volumes of unstructured documents with no automated triage. Led the design and deployment of multi-agent AI systems using LangChain and LangGraph, implementing tool-using agents with memory, task tracking, and goal-directed behavior that autonomously routed findings to compliance and scheduling workflows, cutting manual review effort by 45% and increasing follow-up compliance by 30%.
- Built end-to-end RAG pipelines with chunking, embedding, and vector retrieval layers via SageMaker and Bedrock, with hybrid search and re-ranking achieving grounding levels needed for compliance-critical deployment. Applied prompt engineering, instruction tuning, and LoRA fine-tuning for domain adaptation.

Model Fine-Tuning & Data Curation

- Directed the team's fine-tuning workstream: applied RLHF, instruction tuning, and feedback-driven optimization to align foundation models with domain-specific clinical use cases, overseeing evaluation harnesses covering accuracy, bias, hallucination rate, and user feedback.
- Oversaw data curation pipelines ensuring training datasets were high-quality, diverse, and bias-assessed before any fine-tuning cycle, collaborating with data teams to continuously improve dataset quality based on model evaluation results.

NLP & Classical ML

- Built clinical NLP pipelines using spaCy, scispaCy, BERT, and HuggingFace Transformers for entity extraction, negation handling, and text classification on clinical document corpora, producing structured outputs from high-volume unstructured text.
- Built time-series forecasting models (ARIMA, LSTM, hybrid ensembles) on SageMaker improving forecast accuracy by 18%, reducing claim denials by 25%, and cutting reprocessing costs by 77%. Built segmentation and causal uplift models improving campaign ROI by 25%.

MLOps & LLMOps

- Built CI/CD-driven MLOps and LLMOps infrastructure with Docker, MLflow, SageMaker, Lambda, and Airflow, enabling automated drift detection, retraining, versioning, and rollback across production ML and LLM systems.
- Implemented automated retraining pipelines with Lambda-triggered nightly jobs and S3-based feature storage, keeping production models consistent and safe over time without manual intervention across 25+ sites.

Principal Data Scientist

September 2018 - September 2019

eTrack Tech | Remote

- Built predictive maintenance models across a 10,000+ device IoT fleet reducing fault-driven downtime by 22%, and used NLP to classify root cause patterns from unstructured technician logs, automating a previously manual process.

Graduate Research Assistant

January 2018 - August 2018

Mayo Clinic - Arizona State University Alliance | Arizona

- Performed predictive model evaluation on health services datasets, adhering to IRB-aligned protocols and contributing to data visualization and research reporting.

Product Manager

August 2016 - August 2017

Danfoss | Dubai

- Built financial and risk models for energy solutions, defining product vision and technical roadmaps while partnering with engineering teams to deliver measurable client outcomes.

KEY PROJECTS

Multi-Agent Clinical AI System (Team Lead)

- Manual tracking of findings across high report volumes created compliance risk. Led team design and delivery of a LangChain and LangGraph multi-agent system with DynamoDB-backed memory, tool-calling agents, and secure API integrations, delivered through code review and design sign-off. Increased follow-up compliance by 30%, reduced manual review effort by 45%.

RLHF Fine-Tuning & Evaluation Pipeline

- Production LLM outputs were inconsistent without structured alignment. Directed the team's instruction tuning and RLHF workstream, overseeing prompt engineering strategy, feedback loops, and evaluation frameworks covering factuality, grounding, hallucination rate, and bias detection.

Data Curation & Training Pipeline Infrastructure

- Model quality was limited by inconsistent training data across source systems. Oversaw design and delivery of end-to-end data cleaning, curation, and pipeline infrastructure using Databricks, Spark, Snowflake, and Airflow, with automated quality checks, bias assessment, and schema enforcement feeding model training workstreams.

Scheduling Optimization Framework

- Staffing across 25+ sites was inefficient with no systematic planning tool. Directed delivery of a CP-SAT and genetic algorithm optimization system on AWS Batch with a FastAPI what-if interface, improving staffing efficiency by 35% across all sites.